

Institute of Actuaries of India

Subject CP1– Actuarial Practice (Paper B)

June 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Assignment 1

Solution 1:

Why is pension given to Unorganised workers\self employed?

[2]

This is a group which has very low and irregular income/seasonal income could consist of part time workers/contractual workers/seasonal workers.

Might not have access to level of statutory benefits as organised sector may have.

Standard pension providers like insurance companies would be hesitant to provide pension products offering such low benefits as it does not make economic sense for them.

Neither would insurers be able to offer the type of guarantees that government can offer

Giving access to pension products would enable un-organised sector to have access to affordable pension.

Why entry to be made in the age group 18 -45

[1]

This is the general working life of the unorganised workers during which they can contribute and save.

If workers join beyond age 45, it will be too short a period to accumulate the contributions and may lead to huge deficit/high contributions or need to be funded by the Government.

Further as workers cannot join after 45, it encourages workers to join pension scheme earlier.

Why do contributions increase with age at entry?

[1]

If a worker joins at age 20, he will have 40 years to accumulate his contributions along with government contributions, if any, it can be used to meet the fixed pension per month. Whereas if a worker joins at age 30 he will have only 30 years to accumulate and the accumulated amount can be used to meet the fixed benefit. Hence contribution increases with age at entry.

No penal interest :

[2]

As the group is of unorganised workers, there may be periods when they are without employment and due to the low income level they may be having other commitments to meet and hence not able to contribute to the scheme. This will be a regular feature.

By allowing them to pay without any interest penalty, this scheme ensures that they are covered without a break.

The maximum limit for "contribution holiday" is restricted to 24 months to discourage workers for delaying/discontinuing beyond a limit and the payment with interest may be very high and not affordable for unorganised workers.

Government to make equal contribution upto a limit:

[1½]

Government may consider non income tax paying citizens to be most deserving of government subsidies.

To ensure they have a minimum income post retirement to have a basic standard of living

Encourages workers not to default but save

Commutation and no other withdrawal:

[1]

The workers can commute to meet their immediate needs at age 60. No other withdrawals are allowed as this is a benefit aimed at pension at retirement.

Fixed pension:

A subscriber knows what he is going to get and can Plan accordingly for his retirement. [½]

Benefit to dependant in case of death before 60:

[1]

As the unorganised workers live in poor living conditions and in many cases due to the occupational hazards their mortality may be higher than average population rate.

Hence in case of their death, they may not have any other form of insurance but the dependant will get pension immediately enabling the Government to meet its goal of social security for its citizens.

[8 Marks]

Solution 2:

1. Definition of unorganised worker

[1]

- Definition has to be very clear.
- Even if so, how to check whether the worker applying fits into the definition.
- What will happen if a worker keeps moving from organised to unorganised and vice versa

2. How to ascertain age:

[1]

- What would be age proof.
- Whether any proper age proof will be available.

3. System and Administration

[1½]

- Contribution rate for each member depend on age at entry and pension chosen, the systems should be fully automated and deduction/payment should be from/to the workers bank accounts for success of scheme.
- Unique member ID may be needed which is related to unique government id so that there is no chance of duplicate memberships.
- Link with death registry to stop annuity payments

4. Discontinuance of contribution

[1]

- Mechanism to track contribution at member level has to be there due to discontinuance clause
- Administratively it may not be possible to track them unless unique id is available

5. Dependant pension:

[1]

- Definition of dependant has to be clear.
- Nominations have to be ensured at start to ensure no dispute at settlement

[5 Marks]

Solution 3:i) Factors to consider while setting the contribution rate:

Since it is a government sponsored scheme the general terms will normally be set so that it is more beneficial to the workers. [½]

Risk appetite of government may also be higher than private entities and Government may be more comfortable with lower prudence in assumptions. Hence need to understand risk tolerance and objectives of the Government. [1]

- For the chosen pension, contribution rate have to be decided such that so that the contribution rate is sufficient to meet the value of liabilities. [½]
- Assets are contributions by worker and government plus, Investment returns on investments and [½]
- Liabilities are pensions to member/spouse/dependent [½]

Determination of contribution requires assumptions regarding financial and demographic factors.

Financial Assumptions:

[1½]

Investment Returns which will depend on:

- How the contributions are to be invested.
- Investment strategy.
- Likely Returns in the short, medium and long term on various asset classes
- Riskiness associated with various assets
- Costs of investment management

Expenses:

[1]

Expenses for investment and administrative expenses including checking eligibility, link to check with government registry, collection of contributions, creation and maintenance of IT systems, payment of commutation, annuity to dependant, workers, one off cost of setting up Authority etc to be allowed

Mortality:

[1½]

- Pre 60/post 60 mortality rates to be arrived at both for member
- Mortality rate for spouse/dependents
- Assumption on average age difference between member and spouse
- How to estimate mortality rates of unorganised workers/self employed.
 - To take General population mortality and to estimate expected mortality of the workers.
 - Unorganised workers mortality may be worse than the general population.
- If mortality rates of unorganised workers available it can be taken.

Discontinuance rates:

[1]

- Proportion of workers likely to discontinue the contributions at various durations to be arrived at.
- Contributions paid less expenses could be sufficient/insufficient depending on when the discontinuance happens. Hence this needs to be costed for as well

Other factors:

[1]

As this is a new scheme there is no experience all the assumptions need to be set with margins. There is a significant long-term interest guarantee as well.

However, it should also be ensured that the contribution rates are affordable to the unorganised sector.

[7]

ii) The workers are purchasing lump sum in lieu of the 1/3rd Pension foregone.

In a general scheme, the option terms are *cost neutral* on a given set of assumptions, i.e. to set up an equation of value equating the two forms of benefit (2/3rd pension and lump sum and full pension only option)

The value of pension only option should allow for expected future investment return and post retirement mortality.

It should allow for margins in all assumptions to allow for uncertainty associated with these assumptions.

It is likely that those who are sick are more likely to commute, a decision on whether to allow for higher mortality (which will benefit workers) needs to be considered.

Expenses of paying commutation lump sum should be allowed for

[3]

[10 Marks]

Solution 4:

Options available with the employer

1. “Buy Out” for all and Continue with the current scheme:

[4]

Employer has to buy out of the Government scheme.

Benefits have to match Government scheme

Final minimum monthly pension now has to be more than Rs 5000 for buy out

For 75% of employees benefit has to be increased which would lead to ...

commutation benefit if not offered earlier has to be offered now

---have to match commutation terms offered by Government

---have to offer spouse pension/dependent pension if not offered earlier

---and spouse/dependent pension has to match Government scheme.

All this would lead to

....Sudden increase in scheme liabilities which

...would decrease surplus/increase deficit

....Deficit needs to be funded

....it could be onetime payment to fund deficit

.....or Payment over a few years

2. Buy Out for 25% employees whose benefits are likely higher than Government scheme

[4]

- Contribute to Government scheme for 75% employees from 01.01.2020
- Agreement on transfer value for accrued service
- Continue with existing scheme for other 25% of employees:

For employees to be transferred:

The future contributions will be transferred to the Government scheme. If the current contributions are less than the contribution required by government, the difference has to be funded by employer

Determine “transfer value” for accrued service as at 1-1-2020 for the 75% members

... come to agreement on transfer values which should not impact scheme adversely

... transfer investments or cash value as required by Government

--- if cash has to be transferred assets have to be liquidated which could happen at a loss

--- if assets are to be transferred decide on assets which would need to be transferred for eg., what asset, what credit quality, what duration etc.

For balance 25% of members

Continue with existing scheme

The scheme will suffer from diseconomies of scale due to fewer members than before.

Contribution rates could also go up due to change in age profile and liability profile

There may be change in investment strategy, expenses etc. due to change in liability profile (for eg., all above average aged members may be left changing scheme liabilities)

Trust deed and rules are to be changed and communicated to the employees.

3. Contribute to the Government Scheme for 75% employees and discontinue the scheme for existing employees for new accruals of benefit [3]

..continue the scheme as a closed scheme without further accrual of benefit as it not be economical to run scheme for 25% employees

would cause resentment among these employees leading to attrition

Investment strategy has to be changed as there will be no new contributions to the scheme.

Liquidity issues could arise to make benefit payouts and meet expenses.

Hence assets may have to be realised at unfavourable rates

Higher expenses due to diseconomies of scale and due to effecting the changes

For people with less past service, the pension benefit at retirement may be too low and may not be of much use to pensioners.

The expense of servicing the benefit payout may be too high in relation to the benefit payout.

Other options

(A) Close the scheme for highly paid employees and transfer the fund to the employees on an appropriate basis. This may not be allowed by legislation/trust scheme/rules as it may not serve the purpose of providing pension.

... But there would still need to serve existing pensioners/deferred pensioners. [1]

(B) Close the scheme for the highly paid employees and buy from an insurance company immediate annuity for pensioners and deferred annuity for currently serving employees. [1]

(C) Transfer the liabilities to an insurance company which will guarantee to pay the specified level of benefits.

The insurance company will then accept the risks of future experience.

However, there will be risks for the insurance company in guaranteeing the benefits, for which they will charge a premium.

Hence could be more expensive than managing on own but uncertainty is removed [2]

For the purpose of above, the assets may have to be realised to pay for the benefits. This may lead to assets being realised at most inappropriate time and hence losses. [½]

All options may need the approval of the trustees and (majority of) the employees and changes in the trust deed and rules. [½]

[15 Marks]

Solution 5:

Contract design factors:

Interest and needs of customers:

[2]

Due to the new pension scheme, it appears that many employers with pension schemes may want to purchase these products.

They can purchase single premium/regular premium deferred annuities from insurance company for employees not transferred to government scheme.

- They need immediate annuities from insurance company for their existing pensioners. Hence single premium immediate annuities may be customer requirement
- They need deferred annuity for the deferred pensioners

Some employers may want the annuities to increase in future or some other form of annuity such as joint life etc. The insurance company should have various annuity options in the product design.

Marketability :

[2]

Need to consider innovation to make product marketable like-

- Whether they can purchase the assets directly from the employer instead of making the employer realise the assets and then paying to insurance company.
- Design terms in discussion with investment team for purchase of assets.
- The insurance company should decide in their contract design what would be the death benefit in case of deferred annuity contracts.
- Consider level of sophistication of client/annuitant in understanding the annuity product

Competitiveness:

[1]

Insurer has to ensure that the product does not depart too far from those charged by competitors. Price comparison may be directly available for immediate annuity/deferred annuity contracts. The likely response of competitors to these products should also be analysed.

Risk characteristics:

[1]

The key risks are lower than expected investment returns, increasing longevity and higher than expected expenses. The risk will increase as there will be bulk purchases and the company does not have any past data to price this product. Further the reputational risk could be high if the product is not designed properly and policy wordings are not clear.

Risk appetite and risk aversion:

[2]

These products will have lot of guarantees in respect of interest rate, expense, longevity and that too for a very long period (deferment period and annuity period).

The insurers risk appetite in this regard is very crucial.

The extent to which they are able to match the liabilities with corresponding assets is also an important issue.

General non availability of assets of long duration should also be taken into account.

There could be bulk purchases of annuities in the market which could depress the yield available on the bonds especially long-term bonds.

The insurer can look at options of reinsurance such as longevity swaps can be considered. or offer more life insurance to mitigate this risk.

Profitability:

[1]

As it is a new product, the insurer may not have experience or data to price the product. More margins may be needed in the premium to counter the risk of not meeting the profitability requirement.

The insurer should ensure that the premiums charged will be sufficient to meet the expected annuity payments, meet the cost of guarantee and cost of capital, meet the expected expenses and provide the desired profit margin in most foreseeable circumstances.

There will conflict between profitability, marketability and competitiveness.

Level and form of the benefits:

[1]

May need to offer fixed annuities as well as annuities linked to inflation/subject to flat increase as employer requirements could vary

Some employers may require joint life pension and hence needs to be offered as well

Extent of cross subsidies:

[1]

Extent of cross subsidies between products and within products should be considered. Small sized annuities may need to be offered by insurance companies with some cross subsidy and some expenses of small annuities being met by bigger sized annuities. There will be a risk of the average size of annuities being lower than expected and expenses not being met.

Administrative systems and IT systems:

[1]

The staff should be trained to service these annuities and IT systems are to be purchased/ set up for servicing the deferred and immediate annuities. The expenses are to be spread over the contracts to be sold.

Onerousness of guarantees:

[1]

The insurer should clearly consider the guarantees provided. Annuities and expenses guaranteed for the lifetime of the annuitant. Whether this risk has been properly loaded in the premiums and whether it is marketable after loading for all the foreseeable costs.

Contract terms and conditions:

[½]

The contract terms should be very clear so that there is no ambiguity to the employer and employees.

Financing requirements:

[1]

The financing requirements of annuity and deferred annuity contracts need to be taken into account. As the basic feature is guaranteed the capital requirement could be higher. The feasibility of with profit annuities, not providing guarantee annuities for deferred annuities can be considered.

Trade off between capital intensiveness and meeting needs of customer may be needed.

Regulatory requirement:

[½]

The regulatory requirement needs to be considered in design, pricing, policy documentation, valuing etc of contracts.

[12 Marks]

Assignment 2

Solution 6:

- The basic investment principle of the provider is
 - To maximize overall return on assets where overall return includes both income and capital gains

[½]

- Select investments that are appropriate to nature, term currency and uncertainty of the liabilities [½]
 - Subject to the Risk appetite of the provider [½]
- The annuity liabilities are long term in nature as they are paid till the annuitant is alive [½]
 - The proposed alternative assets may present a good match for the immediate annuities as these are long-term assets [½]
 - The long term bonds may not be available in the market to match the duration of the liabilities in particular corporate bonds [½]
 - Even if they are present they might not be available in sufficient quantities [½]
 - It will help companies to invest in the assets with higher yield subject to its risk taking ability [½]
 - This will enable to enhance the profitability of the portfolio [½]
 - Or to provide competitive rates for the annuity business which is highly competitive in nature [½]
 - There might have been push to look for other asset classes due to significant drop in yields on fixed interest securities [½]
 - The regular income generated from the investments can be used to pay the annuity [½]
 - Since the income produced by assets is likely to increase with inflation it represents good match for indexed linked immediate annuities [½]
 - It will also provide hedge against the expense inflation [½]
 - The immediate annuity cannot be surrendered so they can be backed by illiquid assets [½]
 - The assets considered are illiquid in nature, therefore they are good match to the liabilities [½]
 - There might not be demand for illiquid assets by other investor who may prefer liquid assets in the portfolio [½]
 - They might not be accessible to small investor thereby reducing the competition for the assets [½]
 - Being large also puts the company in a better position to invest in such assets [½]
 - This will further lead to returns which are attractive to annuity provider [½]
 - It will also help in diversification of the asset portfolio as it may have low correlation with the traditional asset classes [½]
 - This may lead to reduction in risk and lower capital requirement [½]
 - Recent changes in the investment regulation may now facilitate investment in such assets [½]
 - They are likely to provide wider investment opportunity set [½]
 - There could be tax benefit associated with certain asset classes [½]
 - Some of these might have lower credit risk if backed by strong institutions like top tier universities [½]
 - They will also provide collateral due to presence of physical assets [½]
 - May also enhance the brand value and image of the provider [½]
 - The competitors are making similar investments to enhance the yield on the asset portfolios [½]
 - Or the company may have a first mover advantage to cherry pick such investments [½]

[15 Marks]

Solution 7:

- Operational risk
 - The company will not have any expertise and experience in assessing and constructing/ managing these kinds of assets which will lead to enhanced operation risks. [1]
 - The skills involved are likely to be substantially different from those needed to manage the other assets in its portfolio. [½]

- Further each of the project is of different nature therefore experience gained might not be directly relevant which leads to increase in operational risk [½]
- The properties may be scattered through a large geographical area leading to further operational difficulties and monitoring risks [½]
- The provider may outsource the operational aspects of managing the assets [½]
- The agreement with outsourcing companies will lead to counterparty risks [½]
- Credit risk
 - There could be issues with regards to honoring of the contracts related to the properties [½]
 - The construction company does not fulfill its contractual obligation like delays in projects, construction not of appropriate quality or leave the construction in between [1]
 - The respective institutions could back-off from the contract due to disagreement over the quality of construction, amenities provided etc. [½]
 - Or the credit quality of the institutions/counterparty has deteriorated post entering into the contract [½]
- Liquidity risk
 - The assets are illiquid in nature, and annuities are illiquid in nature; However liquidity might be required to meet expenses on regular basis [½]
 - Or there could be change in regulation which may introduce surrender values for benefit of the annuitant [½]
 - The transactional costs for selling the assets is likely to be high [½]
- Expense risk
 - There is a risk that the cost of construction and managing these projects is higher than expected [½]
 - This could lead to lower return to the provider and also may lead to strain on capital [½]
 - The inflation of expenses to manage these properties is higher than expected including salary costs [½]
 - There is also the potential for greater than expected cost in terms of diverting management time and other resources away from other management issues [½]
- Regulatory risks
 - There might be changes in the investment regulation which may limit or discourage to invest in such asset classes [½]
 - This may lead to higher capital requirement for investing in such assets [½]
 - Since these assets are illiquid it might be difficult to sell the assets in market if required from any regulatory changes [½]
- Policy risks
 - There could be government policy changes which could favor the tenants for example rent controls etc [½]
 - Given the long-term nature of these investments, there is a risk that a change in government policy reduces the return from the asset [½]
 - Changes in the taxation laws and regulation which may be unfavourable for such investments [½]

- Property risks
 - The provider faces the risks involved in the underlying property assets example: [1½]
 - Obsolescence
 - Void risk
 - Higher dealing and management expenses
 - The provider also faces the risk of physical damage to the developments, for example weather damage, fire etc. [1]
 - Reputational risks
 - The company faces the risk of bad publicity and reputational damage in operating the asset, like evicting the tenants [½]
 - Currency risk
 - Infrastructure investment opportunities may be overseas resulting in the risk of adverse exchange rate movements [½]
 - Market Risks
 - The asset cash flows will not be a perfect match to the annuity cash flows [½]
 - The provider faces the risk of different actual than expected returns on the asset. [½]
 - The assets do not match the liabilities in terms of their exposure to longevity changes [½]
 - Need to hold any mis-match reserve will lead to higher capital requirements [½]
- [16 Marks]**

Solution 8:

- The company should have in place systems to monitor its credit risk exposures [½]
- management information needs to be produced regularly [½]
- There should be well defined risk appetite for credit risk [½]
- Processes should be in place to assess on a continuing basis whether credit exposures are within that appetite [½]
- There could be internal risk committee with appropriate members to discuss credit risk and actions as necessary [½]
- It is particularly important to be aware with the credit risk exposure on each proposed project [½]
- This will enable it to restrict its exposure to different projects and counterparties to levels that are consistent with its risk appetite resources [½]
- However given the nature of investments it will be difficult to put counterparty limit given the nature and uniqueness of each counterparty [½]
- Thorough due diligence should be done before any investments are put in place [½]
- Given that the illiquid nature of these investments will make it hard to disinvest [½]
- It should assess the credit worthiness of its counterparties such as contractors for construction [½]
- To reduce exposure to any one party, the company could diversify the contractors [½]
- It should also consider the replacement risk, i.e.- if one counter party were to default on its obligations [1]
- The company could only enter into projects with institutions having high credit ratings like big universities or multinational companies having sound financials [½]
- It may require collateral or guarantees from these parties to reduce its credit risk exposure [½]
- The company should hire persons with skills needed to assess the credit risks [½]

- Or it could also hire external consultants to assess the risk at initial stage and ongoing basis [½]
- Robust, clearly-worded service level agreements should be put in place with contractors and counterparties [½]
- It could hire external lawyers to vet the key service level agreements and contracts [½]
- The company should be adequately capitalized to commensurate with credit risks taken [½]
- Monitoring credit risk during the term of the project will also be more difficult than for corporate bonds as the infrastructure assets do not have a market price and so there is no observable credit spread. [1]
- This will be made difficult because of the geographical spread and the diversification of the type of investments [½]
- Take insurance if available to insure the credit risk if possible [½]
- It can invest with other investors in large scale projects to enhance the due diligence and use expertise of the other investors [½]

[13 Marks]

Solution 9:

The Incurred but not reported (IBNR) claims means that the death on the annuitant have already occurred but there is delay on reporting the same to the annuity provider. [1]

A key motivation for the dependent not to report the claim is due to continuation of the annuity from the provider till the death is reported. It might be a key source of income for the dependent. [1]

The steps which may be taken by the company to control such risks:

- Annually check the register of death and other databases to see whether the annuitant are is alive or not [½]
- It could ask the annuitant to visit once in a year to a nearby branch/location for validation [½]
- Given that the annuitant might be of old age and due to practicality issues might cause hardship to the annuitant [½]
- This could also lead to reputation issues for the provider [½]
- It could do video authentication for validation [½]
- Ensure it is adequately capitalized [½]
- Ensure adequate allowance for IBNR while pricing the product [½]
- Market fixed period payment annuities, joint life annuities or annuities with death benefit [1]

[6 Marks]
